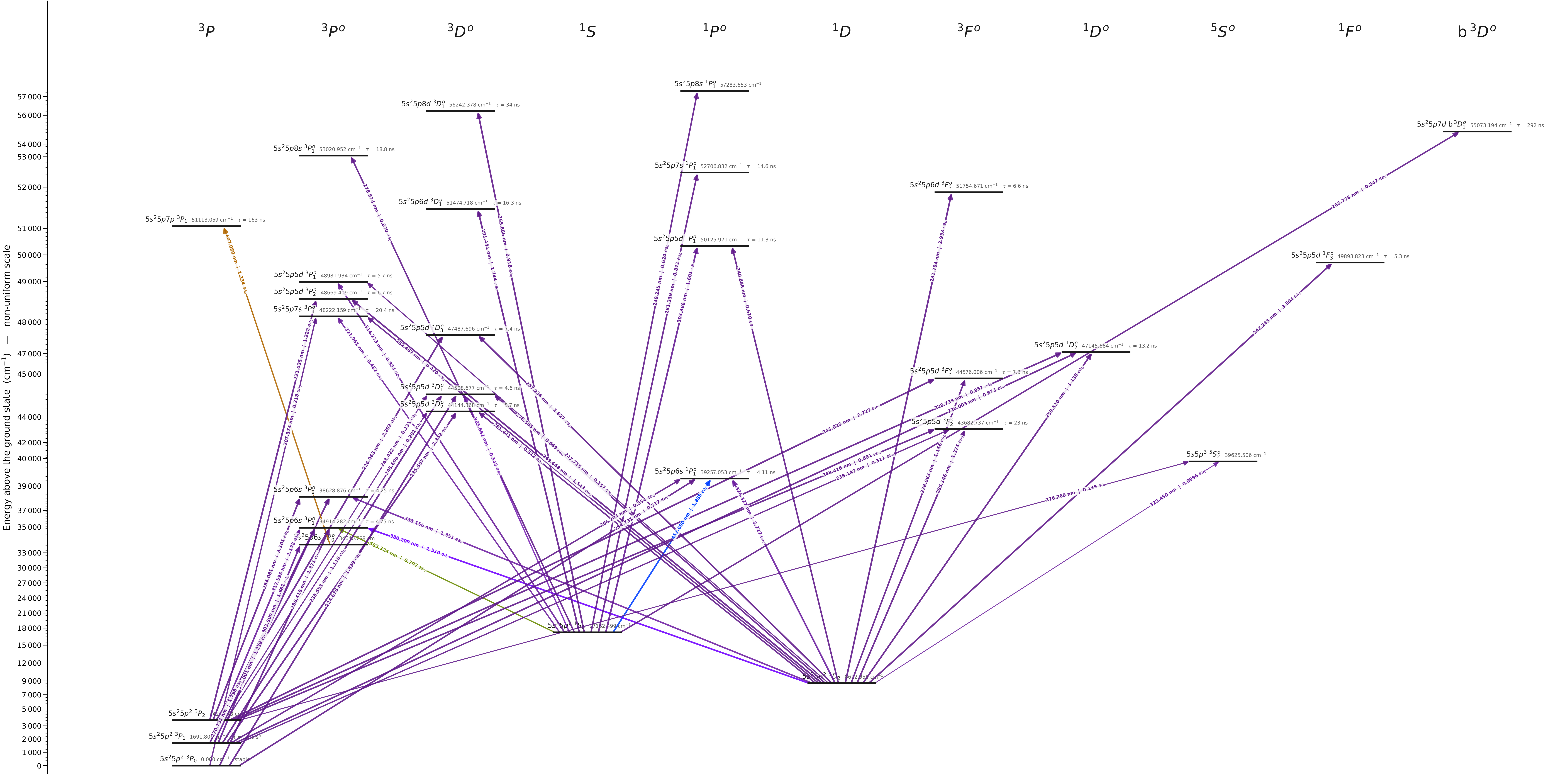
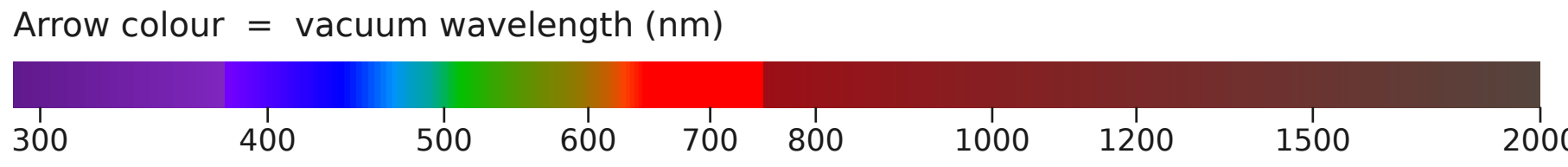


# Tin

Sn (Sn I). Strongest classified transitions below 2 μm (NIST): 29 levels, 51 lines.

Sn<sup>+</sup> ionisation limit 59232.69 cm<sup>-1</sup> = 7.34392 eV (168.826 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|\langle J||er||J'\rangle|$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                                                                                                  | λ vac (nm) | λ air (nm) | ν (THz)   | $ \langle J  er  J'\rangle $ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|-----------------------------------------------------------------------------------------------------------------------------|------------|------------|-----------|-------------------------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s <sup>2</sup> 5p7s <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 281.3388   | 281.2560   | 1065.5923 | 0.871                                           | 2.300e+07            | 10.9           | 33.6       | alternative first step for collinear resonance ionization... |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.0812   | 283.9977   | 1055.3055 | 3.101                                           | 1.700e+08            | 37.45          | 72.2       | first step of the resonance-ionization scheme used for co... |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 286.4157   | 286.3317   | 1046.7038 | 1.371                                           | 5.400e+07            | 33.51          | 25.7       | ground-state resonance line used for high-resolution         |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>1</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 303.4998   | 303.4115   | 987.7847  | 1.661                                           | 2.000e+08            | –              | –          | proposed laser-cooling / magneto-optical-trap transition:... |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 317.5953   | 317.5035   | 943.9448  | 2.178                                           | 1.000e+08            | 33.51          | 47.5       | strongest decay branch of the 5p6s 3P°1 level                |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 452.6002   | 452.4733   | 662.3781  | 1.889                                           | 2.600e+07            | 38.72          | 10.7       | visible line from the metastable 1S0 level used for colli... |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## 119Sn hyperfine structure (*I* = 1/2)

| Level                                                         | A (MHz)      | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|---------------------------------------------------------------|--------------|---------|--------------------------------------------------------|
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>1</sub>   | 578.296(4)   | –       | F=1/2: -578.30 F=3/2: +289.15                          |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>1</sup> D <sub>2</sub>   | -1269.419(3) | –       | F=3/2: +1904.13 F=5/2: -1269.42                        |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>1</sup> D <sub>2</sub>   | -1328.745(4) | –       | F=3/2: +1993.12 F=5/2: -1328.74                        |
| 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | -5011(3)     | –       | F=1/2: +5011.00 F=3/2: -2505.50                        |
| 5s <sup>2</sup> 5p6s <sup>1</sup> P <sub>1</sub> <sup>o</sup> | -259.1(10)   | –       | F=1/2: +259.10 F=3/2: -129.55                          |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                                                                                                  | λ vac (nm) | Isotopes | Shift (MHz) |
|-----------------------------------------------------------------------------------------------------------------------------|------------|----------|-------------|
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s <sup>2</sup> 5p7s <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 281.339    | 116-112  | 491(8)      |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s <sup>2</sup> 5p7s <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 281.339    | 118-112  | 754(6)      |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s <sup>2</sup> 5p7s <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 281.339    | 122-112  | 1126(5)     |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s <sup>2</sup> 5p7s <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 281.339    | 124-112  | 1276(5)     |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.081    | 114-112  | 314(5)      |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.081    | 115-112  | 417.2(22)   |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.081    | 116-112  | 655(5)      |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.081    | 117-112  | 755(4)      |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.081    | 118-112  | 958(3)      |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.081    | 119-112  | 1048(3)     |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.081    | 120-112  | 1225.3(22)  |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.081    | 122-112  | 1460(3)     |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>2</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>2</sub> <sup>o</sup> | 284.081    | 124-112  | 1678(3)     |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>1</sub> <sup>o</sup> | 286.416    | 114-112  | 319.5(16)   |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 286.416    | 115-112  | 412.7(15)   |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 286.416    | 116-112  | 640.0(12)   |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 286.416    | 117-112  | 746.8(12)   |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 286.416    | 118-112  | 945.2(12)   |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 286.416    | 119-112  | 1039.3(15)  |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 286.416    | 120-112  | 1214.4(11)  |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 286.416    | 122-112  | 1448.5(12)  |
| 5s <sup>2</sup> 5p <sup>2</sup> <sup>3</sup> P <sub>0</sub> – 5s <sup>2</sup> 5p6s <sup>3</sup> P <sub>0</sub> <sup>o</sup> | 286.416    | 124-112  | 1656.0(12)  |

Shift = ν(first isotope) – ν(second isotope).